

Diksha Damahe

AI/ML Engineer | +91-9770925370 | dikshadamahe1@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

VIT Bhopal University

2023–2027

B.Tech Computer Science & Engineering (Artificial Intelligence and Machine Learning) | CGPA 8.03

Technical Skills

Languages: Python

Mobile / Web: Dart · TypeScript · Flutter · React · Next.js · PyQt5

Machine Learning: scikit-learn · XGBoost · LightGBM · CatBoost · Isolation Forest · SMOTE · SHAP · SciPy

NLP / DL: Sentence-Transformers (BERT) · TensorFlow · Groq / Llama 3 tool-calling

Databases: SQL · PostgreSQL · SQLite · MongoDB

Serving & data: FastAPI · Django REST · Pandas · NumPy

Tools: Git · GitHub Actions · Vercel · Render · AWS · Jupyter

Experience

Hindustan Copper Limited (Malanjkhand Copper Project)

May 2026 – Jul 2026

AI/ML Engineering Intern

- Built RecovAI on 1,778 digitized shift logs and 26 process variables; XGBoost recovery R^2 0.972 / RMSE 0.139 / MAE 0.104.
- Isolation Forest anomalies, SHAP, PSI drift, SciPy dose search; FastAPI + Groq Llama 3 ReAct agent. Live: hcl-recovai.onrender.com.

Projects

RecovAI — Industrial ML Decision Support

May 2026 – Jul 2026

XGBoost · SHAP · Isolation Forest · FastAPI · Groq · [Live](#) · [GitHub](#)

- Plant-facing recovery predictor, reagent optimizer, anomaly flags, and auto shift reports for Hindustan Copper.

NAFLD Risk Stratification Pipeline

Feb 2026 – Mar 2026

scikit-learn · XGBoost / LightGBM / CatBoost · SMOTE · SHAP · [GitHub](#)

- Merged 6 NHANES 2017–18 files; leakage-safe SMOTE on the training fold only. Benchmarked 24 classifiers; Random Forest 0.9644 test ROC-AUC, 84% sensitivity, 91% specificity.

SPARTA — Phishing Website Detection

2025 – 2026

Python · scikit-learn · IEEE SCEECS 2026 · [DOI](#)

- Ensemble detector on lexical and host-based URL features; 99.15% detection on PhishTank and UCI.

Veil Android

Apr 2026 – Present

Flutter · Dart · ExoPlayer · Riverpod · Hive · [GitHub](#)

- Production Flutter client for Android phone, tablet, and TV: TMDB catalog, REST resolver, ExoPlayer HLS/DASH, Hive persistence, signed CI/CD releases (v3.1.1).

QuantFactor — Factor Research & Backtesting

Aug 2026

Python · pandas · NumPy · SciPy · [GitHub](#)

- Point-in-time backtester with $t+1$ execution lag, commission/spread/slippage, Bonferroni and Benjamini–Hochberg FDR, walk-forward OOS, VaR / CVaR.

Note Assistant — BERT Semantic Search

Mar 2026

Python · Sentence-Transformers · all-MiniLM-L6-v2 · [GitHub](#)

- CLI over personal notes: BERT embeddings, cosine ranking, on-disk cache, 30-question precision / recall / F1 harness.

Publications

SPARTA: Secure Phishing Analysis and Robust Training Algorithm for Phishing Website Detection. Diksha Damahe, Ajay Kumar Phulre. IEEE SCEECS 2026 (Scopus). DOI: [10.1109/SCEECS68810.2026.11430134](https://doi.org/10.1109/SCEECS68810.2026.11430134)

Certifications

Applied Machine Learning in Python — University of Michigan (Nov 2025) · Google IT Support — Google (Jan 2026) · AWS Technical Essentials — Amazon Web Services (Sep 2025) · Cloud Computing — NPTEL, IIT Kharagpur (May 2025) · Introduction to Internet of Things — NPTEL, IIT Kharagpur (May 2026)

Hackathons

Adobe University Hackathon 2026 — Aug 2026 · Women Who Master Hackathon 2026 (Aspire For Her × Logitech) — Jul–Aug 2026 · ET AI Hackathon 2.0 (The Economic Times) — Jun–Jul 2026 · FlowZint AI Hackathon 2026 — May–Jul 2026